

16 September 2026

Dear Editor,

I am submitting for consideration in Nature Machine Intelligence the manuscript "Born wired: innate cortical connectivity plus local plasticity is enough to stand, walk and look" by Zhiwen Li (Independent Researcher, Taicang, Jiangsu, China).

The claim. Machine-learning systems acquire behaviour by optimising a global objective. Brains do not: they are born with a largely pre-specified cortical wiring diagram, and the only rule locally available to a synapse is correlation between the neurons it connects. The manuscript asks how far that combination alone reaches when the cortex has to drive a real body. We built a system of 143,796 model cortical neurons connected by 308,809 pre-specified "instinct" synapses, and it stands up, walks toward a visual target and tracks a moving object with no reward, no error signal, no gradient and no training loop at run time. The one reward-like population in the system is a 200-cell dopamine region; it is used in exactly one experiment and is silent in every other experiment reported.

Why it may interest your readers. The contribution is not a new benchmark score but an existence proof with causal interventions attached, and we think the framing behind it is of interest well beyond this one system: a layered sensory network as a compressor rather than a transducer, thought as co-activation of cortical neurons, the prefrontal population as a second compressor that thinks with the same rules as everything else, and instinct as the initial state of the cortical sheet rather than a module or a set of conditions in code. We state those as six falsifiable claims (P1-P6) and report each result as a test of a specific one.

The interventions are specific rather than correlational. Silencing one population -- the prefrontal one -- takes visually guided walking from 199/200 ticks to 0/200 on 5 of 5 network seeds while leaving reflex-like sense-to-muscle mappings intact. Clearing the instinct table leaves the same five brains unable to stand or approach (0/5). Adding innate structure group by group grows the repertoire without retraining and without destroying earlier capabilities. A capability can also be acquired during the life of one brain: the dopamine population writes one new link from a sound the brain can already hear to an action it can already perform.

Reporting standards. Every number in the paper is produced by a script that is in the public repository, together with the log of the run that produced it. Experiments whose outcome contradicted our prediction are reported as such and collected in their own section (4.5); the manuscript lists, without hedging, what the system does not yet do, including the fact that the acquired walk does not yet stay upright, that the innate wiring is authored by us, and that the locomotion repertoire is distilled from a separately trained controller whose own cost we measured and report (99,483,648 physics steps). The use of AI assistance in producing the code

and the manuscript is disclosed in Section 5.10.

Author situation. This is a single-author, independently funded study by a researcher without a university affiliation who does not write code: the architecture, the six claims, the instinct table, and the requirement that contradicted predictions be reported are the human author's; the code was written in an interactive session with a large language model, and no result was accepted without a run. I mention this because it shapes what I am asking for -- an assessment of the claims, which are stated so that they can fail, rather than of a large group's ongoing programme.

Data and code. All code, logs, figures and reproduction instructions are in the public repository, <https://github.com/leeR1ven/born-wired-cortex> . A four-panel playback page renders the main behaviours as looping animations, so that a reader can watch them without running the simulator.

The manuscript is not under consideration elsewhere and has not been published previously, and no preprint of it has been posted. I declare no competing interests and received no external funding.

Sincerely,

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